Outliers and Skewed Data

What they are, what they do, how to see them, and how to deal with them

About these notes. Illustrated notes for exploratory data analysis and feature preparation. Every idea comes with a picture or a small example, and many use the real columns of the Home Credit loan-default data (307,511 applicants, default rate about 8.07%). The numbers in the project boxes were computed from *application_train.csv*. Rules of thumb (such as the skew bands) are conventions, so treat them as a way to sort your work, not as laws.

How to read the boxes

KEY IDEA
The idea in one or two sentences. If you remember only one thing, remember this.

EXAMPLE
A small worked example with numbers you can redo by hand.

IN YOUR LOAN-DEFAULT PROJECT
Where the idea showed up in the loan-default project, with real numbers.

WATCH OUT
A common mistake or misreading.

The notes in one picture

1 · What Skew and outliers, and where they come from	**2 · Why it matters** Effects on statistics and models	**3 · See and measure** Numbers, rules and charts, incl. log scales	**4 · Deal with it** Fix, cap, transform, or leave alone

The thread running through everything: first **look** (numbers, then charts), then **understand why** a value is extreme, and only then **decide** what to do. Never fix something you have not looked at.

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1 · The big picture

1.1 What is skew?

Skew describes the *shape* of a distribution: whether it is balanced around its centre or has a long tail stretching to one side. A **right-skewed** (positive) distribution has its long tail on the right. A **left-skewed** (negative) one has it on the left. The name comes from the tail, not from where the bulk sits.

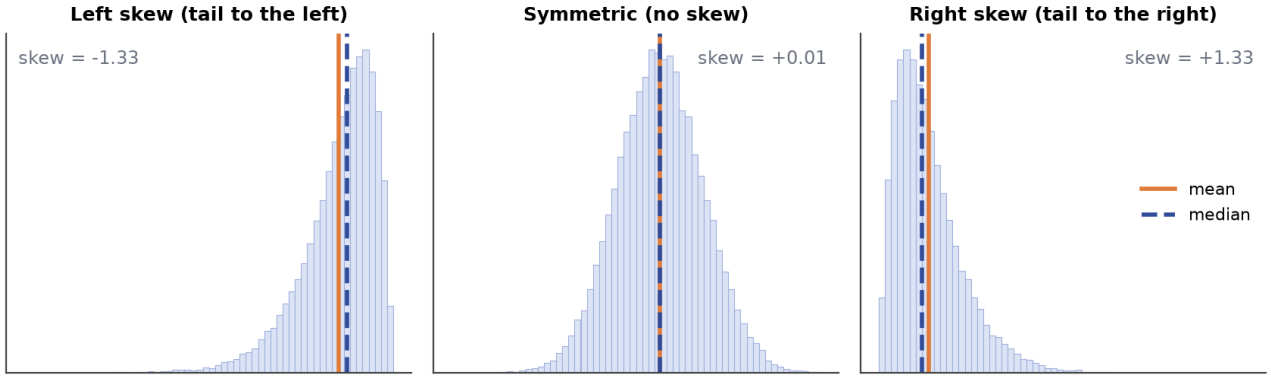


Figure 1.1 · The mean (orange) is dragged toward the tail. The median (blue, dashed) stays with the bulk.

KEY IDEA

In a right-skewed column the **mean is larger than the median**. In a left-skewed column it is smaller. When they are close, the column is roughly symmetric. This is the quickest check you can run in your head.

Skew is calculated from the data, and the number you will see in pandas (**Series.skew()**) is the average of the cubed standardised distances from the mean:

$$\text{skew} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s}\right)^3$$

A value of 0 means symmetric. Cubing keeps the sign, so the tail direction shows. It also gives big weight to far-away points, which matters in section 3.

1.2 What is an outlier?

An **outlier** is a value that sits unusually far from the rest of the data. “Far” is judged against the spread of the other values, not against any fixed number. Two different things get called outliers, and they need different treatment:

	Isolated outlier	Long tail
Picture	One or a few points far from everything else, with a big gap before them	Values thin out gradually; each is not much bigger than the next
Typical cause	Typing error, unit mix-up, placeholder code, one-off event	A natural process (money, counts, waiting times)
Treat it as	Probably an error: investigate, then fix by a rule	Real data: usually transform, not remove
How to tell	Big ratio between the largest and second-largest value	Ratios between consecutive top values are small

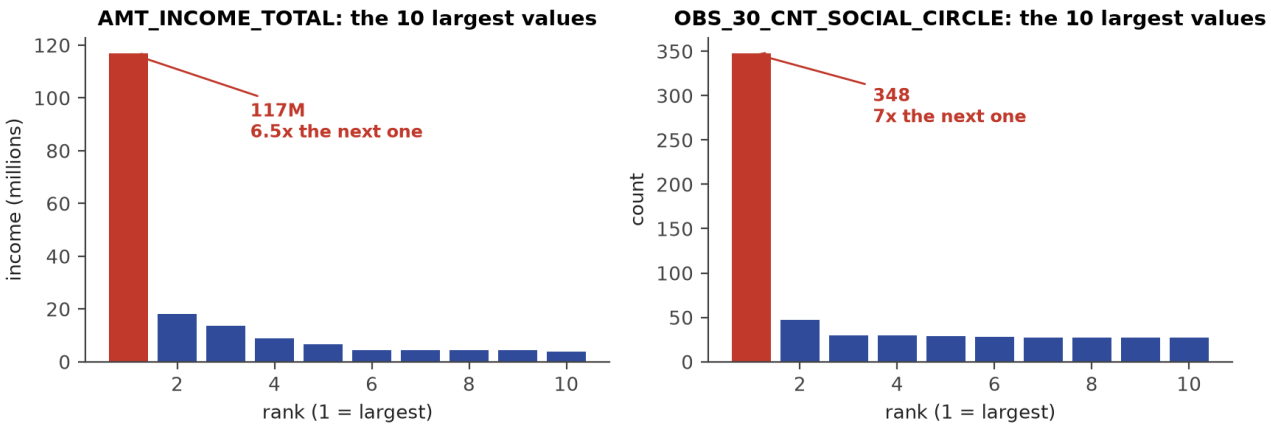


Figure 1.2 · Both columns have an isolated top value, far above the rest. Compare with a genuine long tail below.

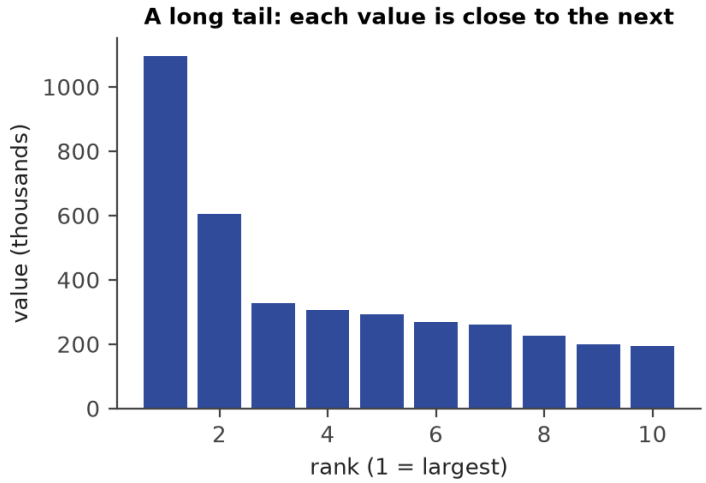


Figure 1.3 · A long tail: the top values step down smoothly. Nothing here looks like a mistake.

IN YOUR LOAN-DEFAULT PROJECT

AMT_INCOME_TOTAL: the largest income is 117,000,000, then 18,000,090, then 13,500,000. The first is 6.5x the second, an isolated value that is almost certainly an entry error (it sits next to a 562,491 loan). **OBS_30_CNT_SOCIAL_CIRCLE:** the top values are 348, 47, 30, 30. Again one value (348) stands apart.

1.3 Skew and outliers are related, but not the same

They often appear together, because a few far-away points create a tail. But you can have one without the other.

Situation	Skew	Outliers
A smooth income-like column	High	Few real errors, just a long tail
A symmetric column with one typo (age 900)	Can still be large because of the one point	One clear outlier
A column with a placeholder spike (365243)	Large, but caused by the placeholder	A whole block of fake values
A bounded 0 to 1 column with a pile near 0	Moderate to high	None, the tail is real and bounded

WATCH OUT

A high skew number does **not** prove there are outliers, and a clean-looking column can still hide one. Always look at both the number and the chart.

1.4 Where they come from

Source	What happens	Example
Natural, multiplicative processes	Quantities grow by percentages, so the top is far above the bulk	Income, loan size, house prices
A floor at zero	Values cannot go below 0, but can go far above	Counts, amounts, durations
Rare events	Most rows have 0, a few have many	Credit enquiries, defaults in the social circle
Data entry errors	Wrong digit, extra zero, wrong unit	An income of 117,000,000
Placeholder (sentinel) codes	A fake value standing for “unknown”	DAYS_EMPLOYED = 365243 (about 1000 years)
Mixed populations	Two groups with very different scales in one column	Pensioners vs employed for employment length

Check yourself (answers in Appendix C)

- Which is larger in a right-skewed column, the mean or the median?
- How can you tell an isolated outlier from a long tail using only the top few values?
- Give one example of skew with no outliers and one of an outlier that is not a long tail.

2 · What they do to your analysis

2.1 Effects on simple statistics

Some statistics are **sensitive** (a single point can move them a lot). Others are **robust** (they hardly react).

Sensitive to extremes	Robust to extremes
Mean	Median
Standard deviation and variance	IQR, MAD (median absolute deviation)
Skew number, Pearson correlation	Spearman (rank) correlation, quantiles
Min and max, range	Percentiles such as p5 and p95

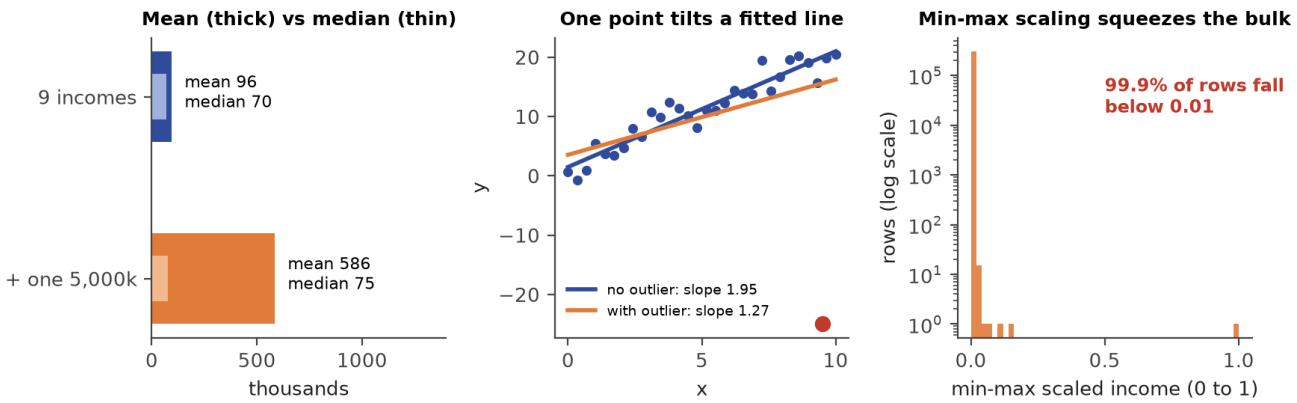


Figure 2.1 · Left: nine incomes, then one extra 5,000k. Middle: one point tilts a regression line. Right: min-max scaling puts 99.9% of the rows in the first sliver of the 0 to 1 range.

EXAMPLE

Nine incomes (in thousands): 40, 50, 55, 60, 70, 80, 90, 120, 300. The mean is 96 and the median is 70. Add one income of 5,000. The mean jumps to 586, but the median only moves to 75. One row changed the mean six-fold.

2.2 Effects on models

Model family	How it reacts to skew and outliers in the features	Usual consequence
Linear and logistic regression	Squared (or log) loss gives far points huge influence (leverage). Coefficients assume a roughly linear effect.	A few rows dominate the fit; coefficients become unstable
Neural networks	Large inputs saturate activations and make training unstable unless scaled	Slow or erratic training
k-NN, SVM, clustering	Distances are dominated by the column with the largest scale or spread	One wide column decides who the neighbours are
Scaling (min-max, z-score)	Min-max depends on the extreme values; the standard deviation is inflated by outliers	The bulk is squeezed into a tiny range
Trees, random forest, boosting	Splits use the *order* of values, not their size, so a monotonic change such as log changes nothing	Mostly unaffected by feature outliers
Any model on a skewed *target*	Squared-error training chases the largest target values	Poor fit on the typical case

KEY IDEA

Trees care about the **ranking** of values. Linear models, neural nets and distance-based methods care about the **actual numbers**. That single difference decides most of the “do I need to fix this?” questions in section 6.

IN YOUR LOAN-DEFAULT PROJECT

In your own numbers: **AMT_INCOME_TOTAL** has a skew of **391.6**. Remove the single 117,000,000 row and it is still **27.0** (the next values 18M, 13.5M, 9M keep it high). Take log10 of the whole column and it drops to **0.17**. One transform did what deleting a few rows could not.

Step	AMT_INCOME_TOTAL skew
Raw column	391.6
Without the 117M row	27.0
log10 of the raw column	0.17

Check yourself (answers in Appendix C)

- Why does a single extreme value affect the mean but hardly move the median?
- Which model family is least affected by an outlier in a feature, and why?
- Why is min-max scaling risky on a column with an outlier?

3 · How to measure them

3.1 The skew number and its bands

Use the skew number to **sort** your columns, then confirm with a chart. These cut-offs are conventions:

|skew|	Reading	Typical action
below 0.5	Roughly symmetric	Nothing to do
0.5 to 1	Moderate skew	Look at the plot; often fine as it is
above 1	Strong skew	Consider a log or similar transform
above 2 or 3	Extreme: a long tail or outliers	Investigate the tail first, then transform

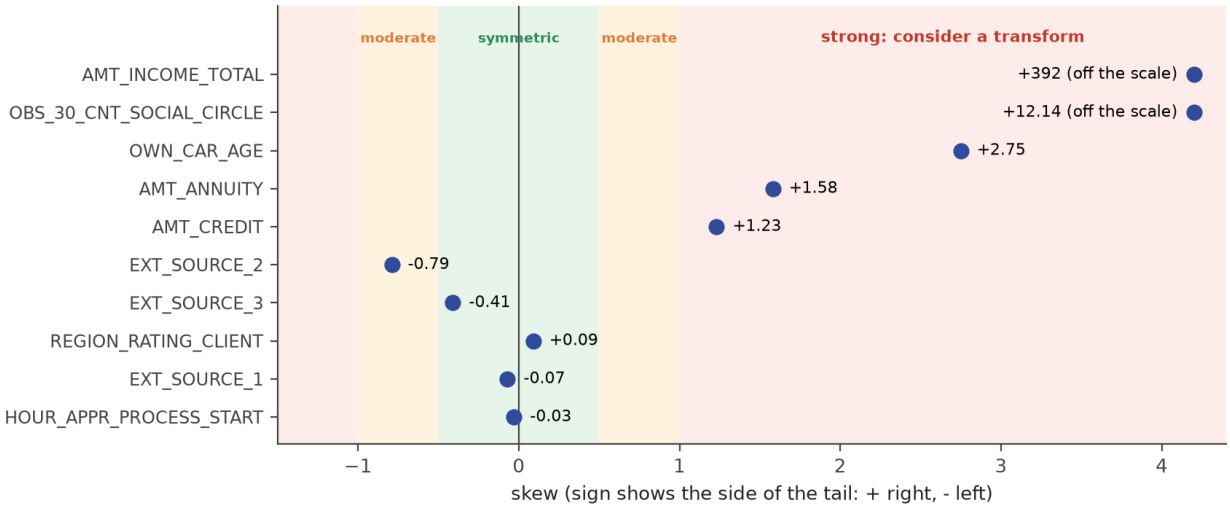


Figure 3.1 · Skew of columns in the loan-default data, sorted by size. Shading shows the bands.

WATCH OUT

Read the absolute value. A skew of −0.9 is moderate, not “low”. EXT_SOURCE_2 sits at −0.79, so it is moderately left-skewed, not symmetric. Also remember the sign flips if you flip the axis: DAYS_EMPLOYED stored as negative days looks left-skewed, but after converting to positive years it is right-skewed.

When the skew number misleads

- A **single extreme point** can create a huge skew (as with income at 391.6) while the bulk is perfectly fine.
- A column with a **big spike at one value** (zeros, a placeholder) has a skew that says little about the shape of the rest.
- For columns with **very few distinct values**, skew is poorly defined. Use value counts and a bar chart.

3.2 Three quick checks to tell moderate skew from none

Check	How	Symmetric if…
Mean vs median	(mean − median) / std	close to 0 (say within ±0.1)
Percentile spacing	compare p50 − p5 with p95 − p50	the two distances are about equal
Before and after a log	skew of log1p(column) vs skew of the column	it barely changes (if it drops sharply, the skew was real)

$$\frac{\bar{x} - \text{median}}{s}$$

A standardised gap between mean and median. Positive means right-skewed.

3.3 Rules to flag outliers

No rule is perfect. Use two different ones and look at what both flag.

The ratio screen (fast, good for finding isolated values)

- **max / p99**: if the maximum is far above the 99th percentile, the top 1% is stretched.
- **max / second-largest**: if one value is several times the next, it is isolated, as with 117M against 18M.
- Division by zero gives **inf** when the denominator is 0. That is a sign the column is mostly zeros, not that something is broken.

The IQR rule (the box plot rule)

$$\text{IQR} = Q_3 - Q_1, \qquad \text{fences} = [\, Q_1 - 1.5\,\text{IQR}, \quad Q_3 + 1.5\,\text{IQR}\,]$$

Anything beyond the fences is drawn as a dot on a box plot. It is robust, because the quartiles ignore the extremes. The 1.5 is a convention; for a heavily skewed column the rule flags thousands of ordinary rows on the long side, so on skewed data apply it *after* a log transform.

The z-score rule

$$z = \frac{x - \bar{x}}{s}$$

Often flags |z| above 3. It works for roughly bell-shaped data. But outliers inflate the standard deviation, which shrinks their own z-score (masking).

The robust z-score (median and MAD)

$$z_{\text{robust}} = 0.6745\,\frac{x - \text{median}}{\text{MAD}}, \qquad \text{MAD} = \text{median}(|x_i - \text{median}|)$$

Median and MAD barely move when outliers are present, so the outliers stand out clearly.

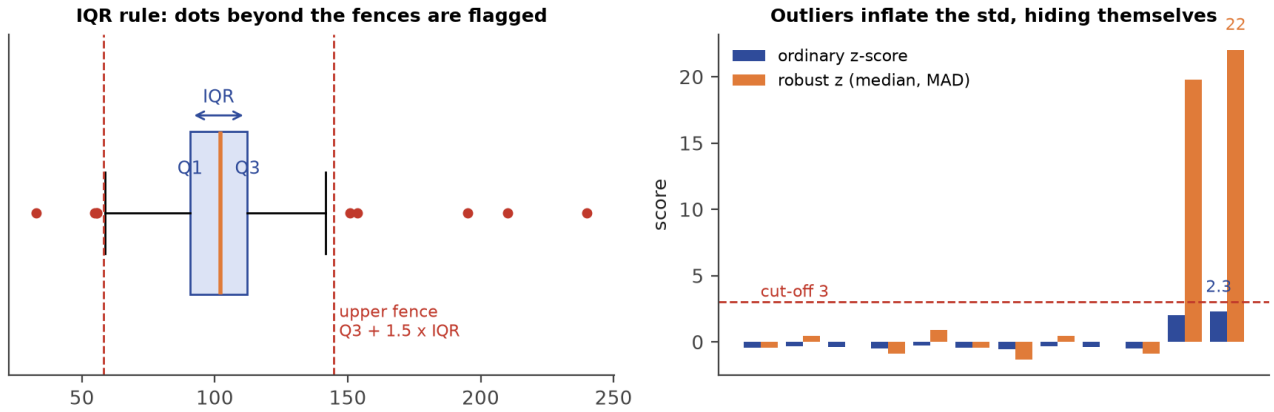


Figure 3.2 · Left: the IQR rule on a box plot. Right: with two extreme points, the ordinary z-scores are only about 2, so the 3-cut-off misses them. The robust z-scores are near 20.

Rule	Good for	Weak spot
max / p99, max / second	A quick scan of every column	Not a yes/no verdict; needs a look
IQR fences	Box plots, moderately skewed data	Flags too many rows on long tails (do it after a log)
z-score above 3	Roughly bell-shaped data	Masking: outliers hide themselves
Robust z (MAD)	When you suspect several outliers	Fails if more than half the values are the same (MAD = 0)

KEY IDEA

A statistical rule only says a value is **unusual**. Whether it is **wrong** depends on the real-world meaning (can a person have 348 enquiries? can someone earn 117M and take a 562k loan?).

Check yourself (answers in Appendix C)

- EXT_SOURCE_2 has a skew of −0.79. Which band is that, and which side is the tail?
- Why can the ordinary z-score miss an outlier?
- Why should you use the IQR rule after a log transform on a long-tailed column?

4 · How to view them

4.1 Six views of the same column

The same column (applicant income, without the single 117M row) shown six ways. Each view answers a different question.

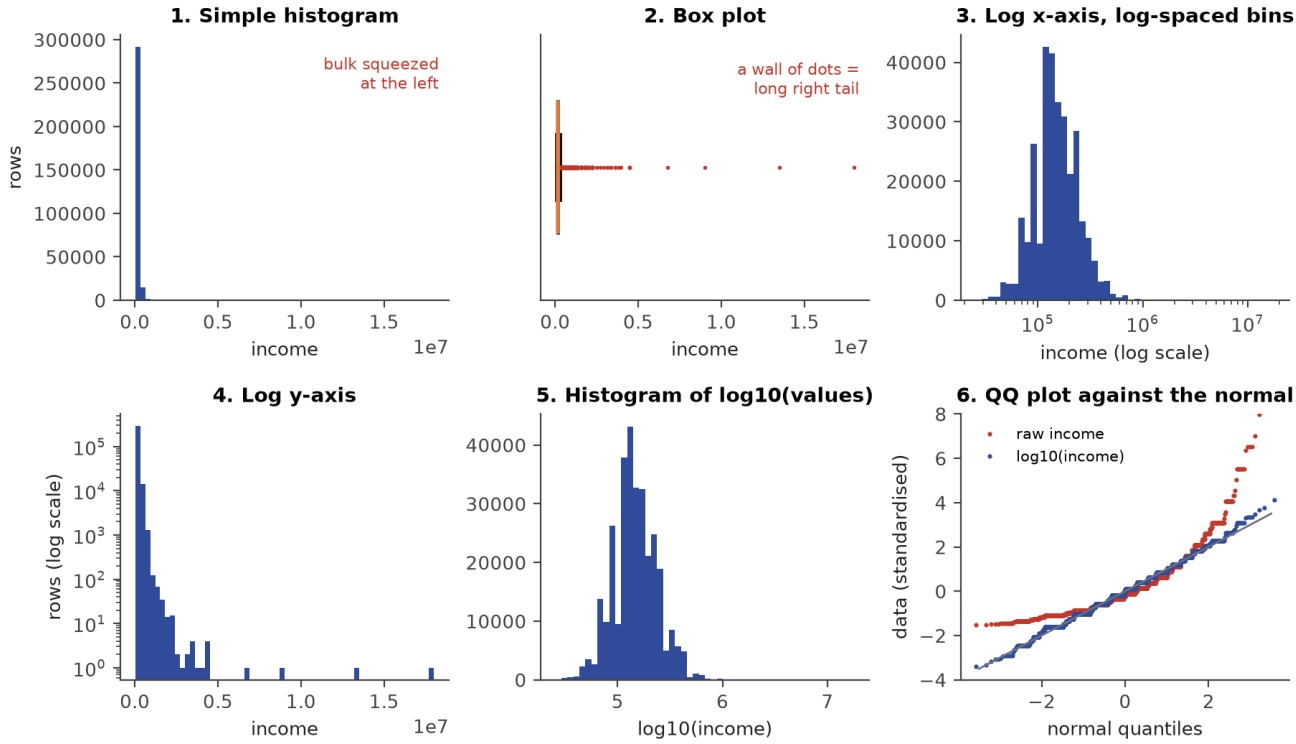


Figure 4.1 · AMT_INCOME_TOTAL under six views.

View	Answers	Weak spot
1. Simple histogram	What does the bulk look like in real units?	A long tail squeezes everything into one bar
2. Box plot	Where are the quartiles, and how many points are beyond the fences?	Hides the shape (bimodal, spikes)
3. Log x-axis histogram	What is the shape across orders of magnitude?	Needs positive values; counts per bin are not density
4. Log y-axis histogram	Does a thin tail exist beneath a tall bar?	Easy to overread small bars
5. Histogram of log10(values)	What will a model see after a log transform?	Axis shows log numbers, not real units
6. QQ plot	Is the column close to normal? Where does it depart?	Takes practice to read

How to read a QQ plot

A QQ plot sorts your data and plots each value against where it *would* fall if the data were normal. If the points hug the diagonal line, the column is close to normal. A curve that bends up at the right end means a heavier right tail than the normal. In Figure 4.1 the raw income (red) bends sharply away, while log10(income) (blue) follows the line closely.

IN YOUR LOAN-DEFAULT PROJECT

In your plan, every numeric column should end in one of five groups: log x-axis histogram, log y-axis histogram, histogram with box plot, bar chart (few distinct values), or a days-to-years histogram. The table in section 7 shows the mapping.

4.2 Why a log scale

A normal axis spaces values by **difference** (0, 1,000k, 2,000k). A log axis spaces them by **ratio** (each step is twice the last). Money behaves by ratio: the gap between 40k and 80k matters as much as between 1M and 2M. So a log axis shows each of these gaps with equal width, and spreads out the crowded left side.

EXAMPLE

Ten incomes (thousands): 40, 50, 55, 60, 70, 80, 90, 120, 300, 5000. With equal-width bins of 1000 the first bin holds 9 of the 10. With doubling bins (25 to 50, 50 to 100, 100 to 200 …) the counts are 2, 5, 1, 1, 1 (and 1 in the far bin): now you can see a peak and a thin tail.

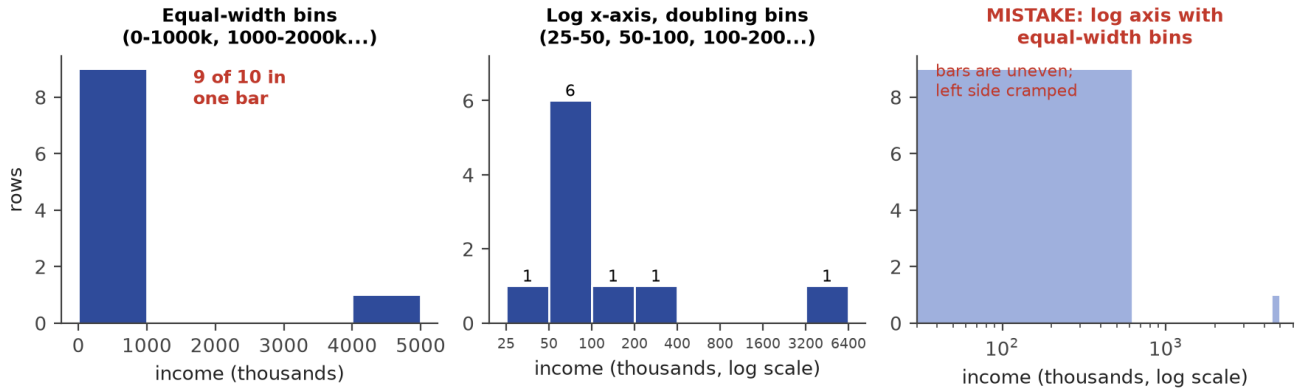


Figure 4.2 · Left: equal bins hide the shape. Middle: doubling bins on a log axis show it. Right: the common mistake, a log axis with equal-width bins.

What the axes hold. In a log-x histogram the x-axis still shows *values* (labelled in real units) and the y-axis shows *counts*. Only the spacing of the bins changes: bins are narrow on the left and wide on the right, so each bin spans the same *ratio*. Because the bins are wider at high values, bar height is a row count, not a density.

WATCH OUT

Setting a log x-axis is not enough. You must also use **log-spaced bins** (**np.logspace**). With equal-width bins the bars look uneven and the left side is cramped (Figure 4.2, right).

4.3 Five types of log histogram

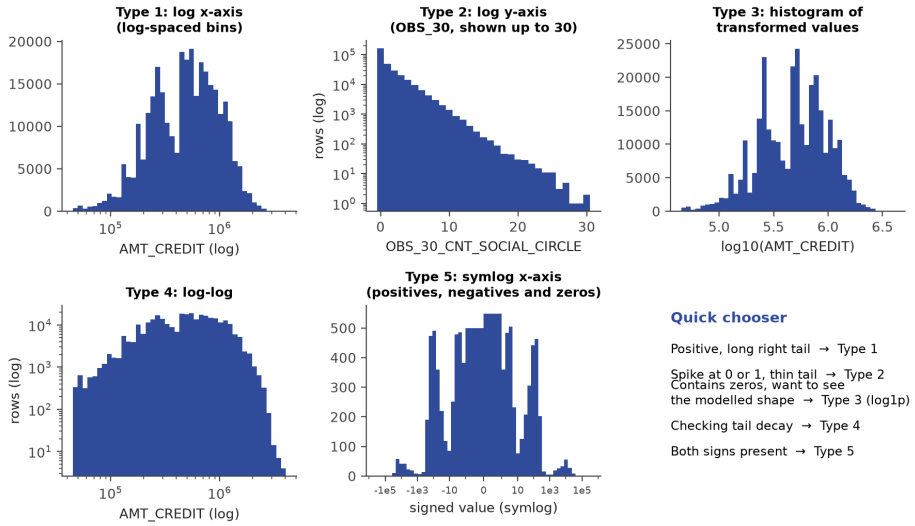


Figure 4.3 · The five types, plus a quick chooser.

Type	What is logged	Use when	Zeros OK?
1. Log x-axis, log-spaced bins	The values (x)	Positive, right-skewed values (AMT_*)	No
2. Log y-axis	The counts (y)	A few huge bars hide a thin tail (OBS_*, housing columns)	Yes
3. Histogram of transformed values	The values, before plotting	You want the shape after the modelling transform	log1p keeps zeros
4. Log-log	Both axes	Checking whether the tail decays regularly	No
5. Symlog x-axis	Linear near 0, log far from it	Both positive and negative values	Yes

Rule of thumb: if the problem is a long tail of *large values*, change the x-axis. If the problem is a few *dominant bars*, change the y-axis. For a column with zeros, a log x-axis cannot show them, so use a log y-axis or a log1p transform.

WATCH OUT

Reading a log y-axis. Equal vertical distances mean equal *ratios*, not equal differences. A bar that looks half as tall can hold 100 times fewer rows. Use it to check that a tail exists, and write the real counts or shares in your notes.

IN YOUR LOAN-DEFAULT PROJECT

Which type for which column. Money and positive tails (AMT_INCOME_TOTAL, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE, REGION_POPULATION_RELATIVE): type 1. OBS_30/60_CNT_SOCIAL_CIRCLE and bounded 0 to 1 housing columns with many zeros: type 2 (for integer columns use integer bin edges). Columns with zeros where you want to see the modelled shape (OWN_CAR_AGE): type 3 with log1p. Types 4 and 5 are rarely needed here.

Check yourself (answers in Appendix C)

- In a log-x histogram, what is on each axis, and what changes compared with a normal histogram?
- Why does a log x-axis fail for OBS_30_CNT_SOCIAL_CIRCLE, and what do you use instead?
- What does it mean if the raw QQ plot bends upward at the right end?

5 · How to deal with them

5.1 Decide first: what kind of extreme value is it?

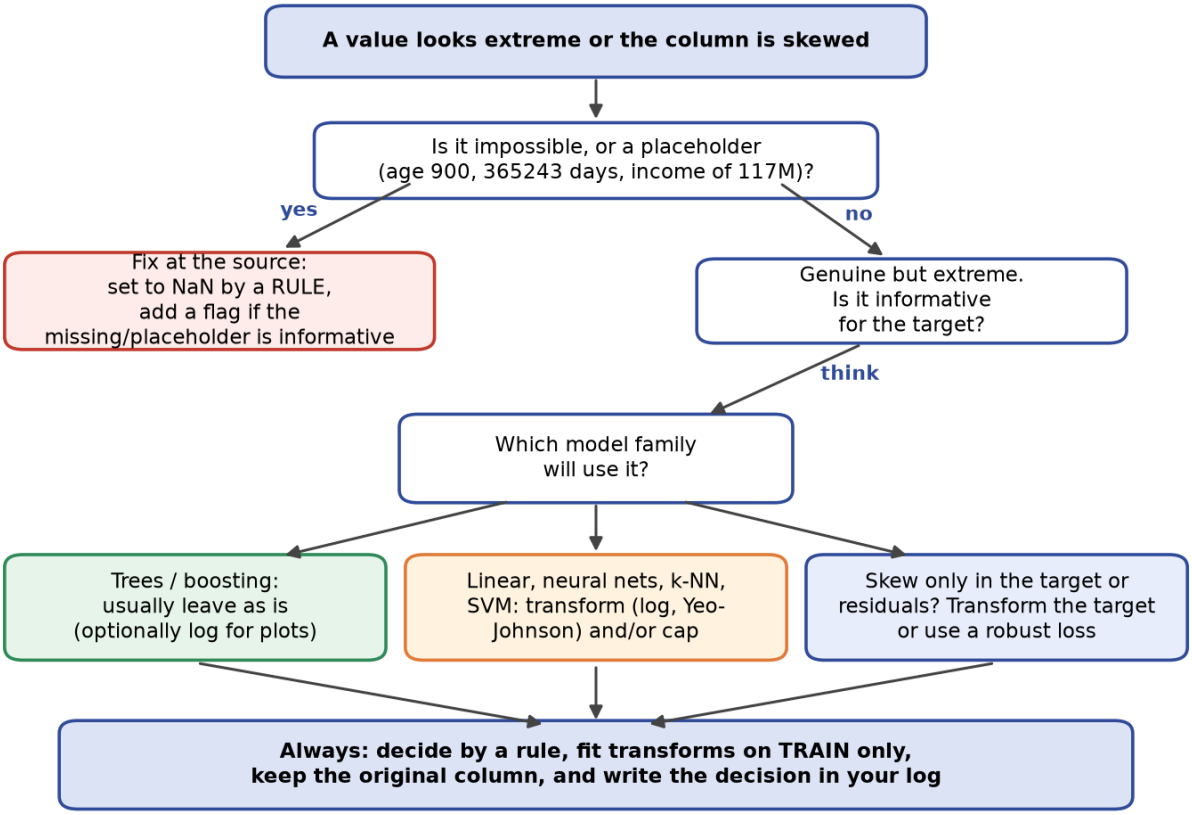


Figure 5.1 · A decision path. The first question (error or genuine?) matters most.

The order matters: first **fix what is wrong** (errors and placeholders), then **decide what to do with what is real** (transform, cap or leave). Never transform or cap a column that still contains placeholders, or the fake values will distort the result.

5.2 The toolbox

Option	What it does	Use when	Watch out for
Fix by a rule	Set impossible values to NaN (e.g. count above a sane limit)	The value is an error or placeholder	Use a rule on the value, never a row number
Add a flag	A 0/1 column “was a placeholder/missing”	The fact it was extreme or missing carries signal	Check the flag against the target first
Cap (winsorise)	Clip values above p99 (and below p1)	Real values, but a few are extreme; linear or neural models	Thresholds must come from the training data
Log or log1p	Compress large values	Positive, right-skewed, multiplicative data	log needs x > 0; log1p handles zeros; can overshoot to left skew
Square root	Milder compression	Counts, moderate skew	Weaker than log
Box-Cox / Yeo-Johnson	Finds the best power for you	Strong skew, want a near-normal shape	Box-Cox needs x > 0; Yeo-Johnson allows zeros and negatives
Rank / quantile transform	Replaces values by their rank, mapped to normal	Heavy tails you cannot fix otherwise	Loses the distances between values
Binning	Groups values into ranges	Tree-friendly or interpretable features	Throws away detail inside a bin
Robust scaling	Scale by median and IQR instead of mean and std	Outliers remain, model needs scaled inputs	Does not change the shape
Robust loss / model	Huber loss, quantile regression, trees	Outliers in the target	Changes what the model optimises
Drop the row	Remove the observation	Last resort for a proven error with no fix	Loses data; can bias the sample
Leave it	Do nothing, document why	Tree models on a genuine value	Write the decision in your log

$$y = \log(1 + x)$$

$$y = \frac{x^{\lambda} - 1}{\lambda} \quad (\lambda \neq 0), \qquad y = \log x \quad (\lambda = 0)$$

Box-Cox: the lambda is chosen to make the result as close to normal as possible.

$$x_{\text{capped}} = \min(\max(x,\ p_1),\ p_{99})$$

5.3 Transforms, side by side

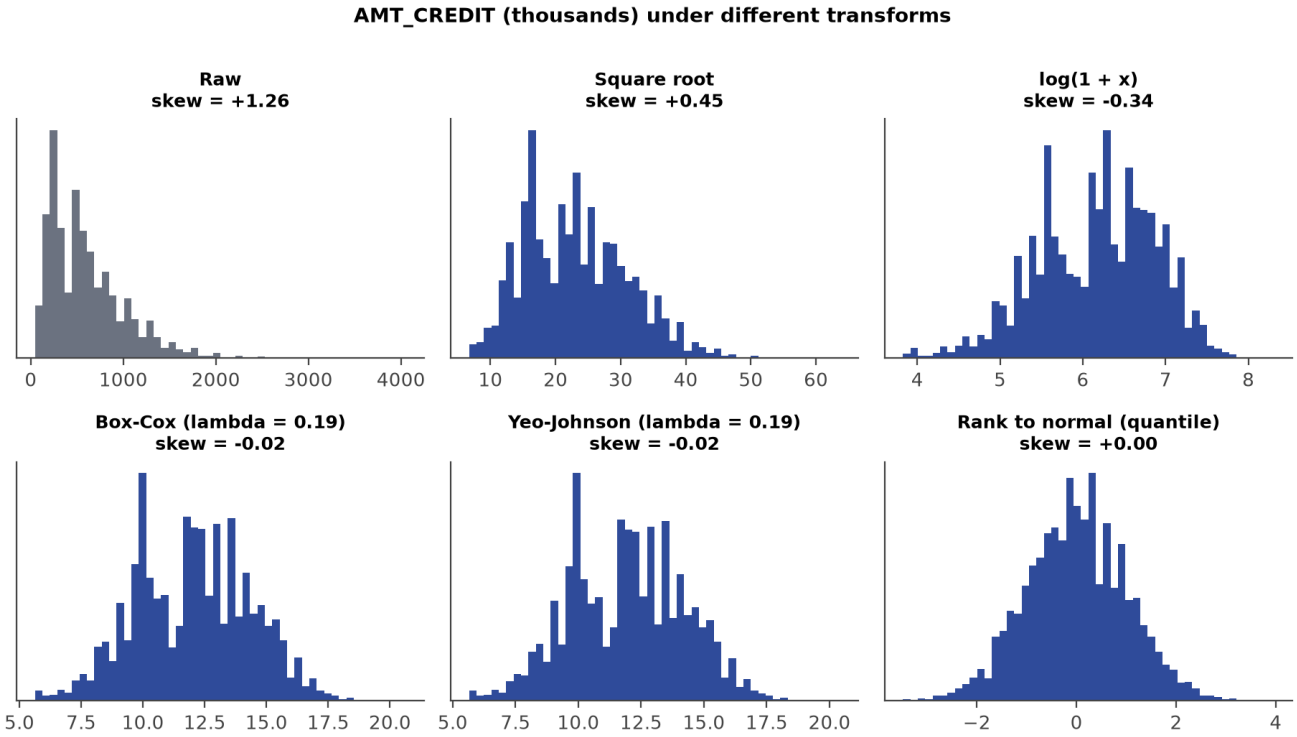


Figure 5.2 · One column under six transforms (AMT_CREDIT, thousands).

- The **square root** and **log** both reduce the skew. Here the log *overshoots* to −0.34 (a mild left skew), so more is not always better.
- **Box-Cox** and **Yeo-Johnson** pick lambda (0.19 here, close to a log). They land on a skew near 0.
- The **rank transform** forces a normal shape exactly, but throws away the real gaps between values.
- A transform changes the *shape* only. Spikes at round values (loan products) stay, because they are real.

> **KEY IDEA**
> A transform is a tool to meet a model's needs, not a cleaning step. If the model does not care (trees), you may not need one.

5.4 Capping (winsorising)

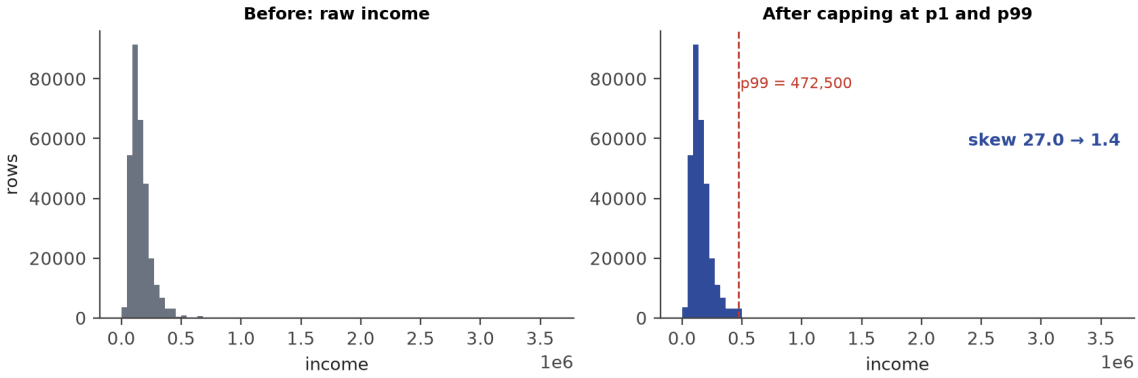


Figure 5.3 · Capping at the 1st and 99th percentile pulls the far tail in and reduces skew sharply.

Capping keeps every row but limits how far any single value can pull. Use it when the extreme values are real (so you do not want to delete them) but would dominate a linear model or a distance-based method. Choose the percentiles from the **training** data and apply the same limits to validation and test data.

WATCH OUT

Capping hides information: after capping, a 5M income and a 900k income look the same. If extreme values might matter to the target, test the capped and uncapped versions.

Capping versus winsorising

Both replace values beyond a limit by the limit. The only difference is **where the limit comes from**. Winsorising is a special case of capping, so the code is usually the same **clip** call.

	Capping	Winsorising
Limit comes from	Your choice: a fixed number from domain sense or from the plot	A percentile of the data (for example p1 and p99)
Example	DEF_30_CNT_SOCIAL_CIRCLE capped at 5, so 34 becomes 5	AMT_INCOME_TOTAL clipped to the training p1 and p99
Good for	Counts with a meaningful maximum, known business rules	Continuous columns with no natural limit
Learned from data?	No, if the number is hand-picked	Yes: compute percentiles on the training set only

```
# capping: a hand-picked limit
df['DEF_30'] = df['DEF_30'].clip(upper=5)

# winsorising: limits learned on train only
lo, hi = train['AMT_INCOME_TOTAL'].quantile([0.01, 0.99])
df['AMT_INCOME_TOTAL'] = df['AMT_INCOME_TOTAL'].clip(lo, hi)
```

5.5 Placeholders: the cleanest fix

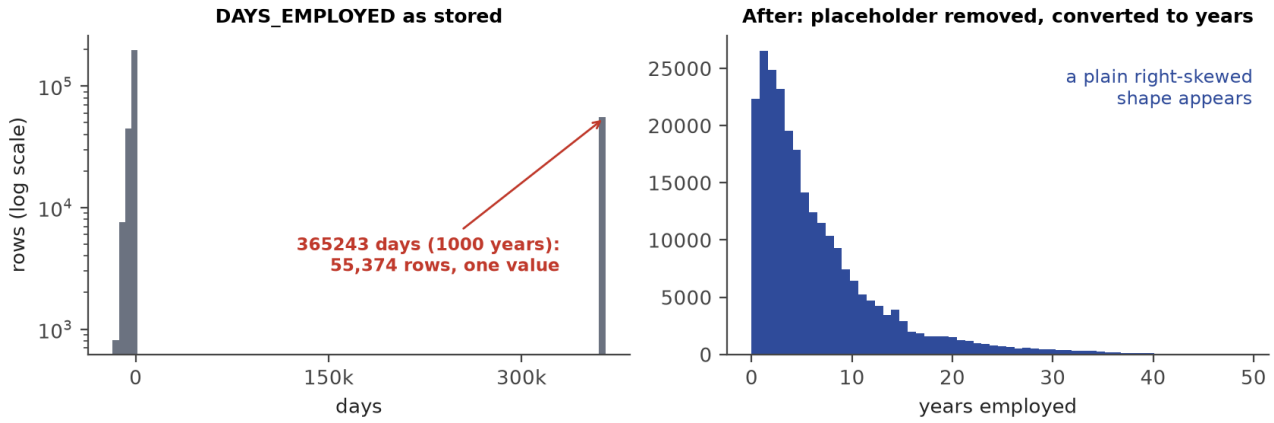


Figure 5.4 · DAYS_EMPLOYED holds a block of 55,374 rows at 365,243 days. Remove the placeholder and convert to years.

A placeholder is not an outlier to transform. It is a **code for "unknown"**. Replace it with a proper missing value by a rule (**value == 365243**), and consider a flag, because who gets the placeholder is often informative (here these are mostly pensioners). Only then look at the real distribution.

IN YOUR LOAN-DEFAULT PROJECT

DAYS_EMPLOYED: 55,374 rows (about 18%) equal 365243. After removing them and converting days to years, the remaining column is a plain right-skewed shape. The same logic applies to the AMT_REQ_CREDIT_BUREAU_QRT = 261, the OBS_30 = 348 and the 117M income: set them to NaN by a rule on the value, log each, and test an imputed value plus a flag.

5.6 Skew in the target

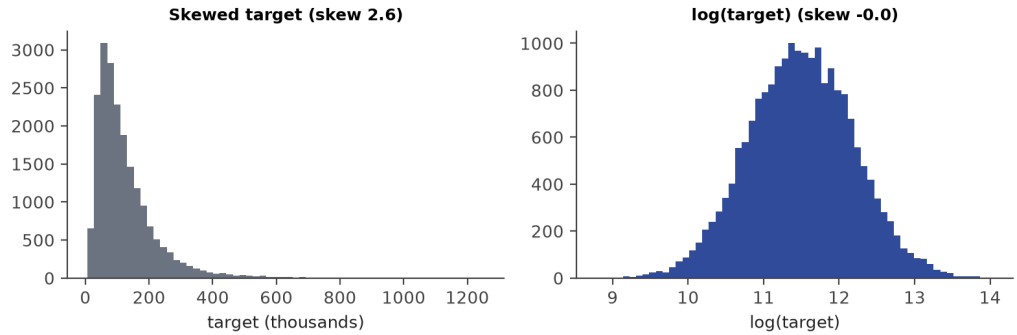


Figure 5.5 · A skewed target (such as a loan amount) after a log transform.

If you were predicting a continuous, skewed target, squared-error training chases the largest values. Common fixes: model **log(target)** and convert back, use a Huber or quantile loss, or use a tree model. Your project target (default or not) is binary, so this applies only if you later add a regression task.

5.7 Match the tool to the pattern

The first job is to **name the pattern**. Each pattern has its own tool, and a tool used on the wrong pattern does little or harm. Outliers are usually capped or winsorised; a long tail is usually log-transformed.

Pattern	How to tell	Usual tool	Not the right tool
Isolated outlier	Max is far above the second-largest (117M vs 18M)	Inspect the row, fix if an error, otherwise cap or winsorise	Log alone: the point stays far away
Long tail	Gradual staircase of large values, skew above 1	Log or log1p; sqrt; Yeo-Johnson	Capping: it cuts many real values
Zero pile	Large share of exact zeros (zero % above about 30)	“Zero or not” flag	Log or cap: the spike stays
Placeholder	Impossible repeated value (365243)	Set to NaN by a rule, add a flag	Any transform before cleaning
Wall at the maximum	Spike at a bound such as 1.0 on a 0-to-1 column	Usually leave alone	Treating it as an outlier
Plot artifact	Gaps only on one axis type (OWN_CAR_AGE on log bins)	Replot with integer bins	Changing the data

Other options for outliers and long tails

Situation	Options
Isolated outlier	Fix the value if it is an error; cap or winsorise; drop the row if proven corrupt; flag then cap; do nothing for trees
Long tail	log / log1p; square root (milder, counts); Yeo-Johnson or Box-Cox (learned on train); do nothing for trees

IN YOUR LOAN-DEFAULT PROJECT

AMT_INCOME_TOTAL has both a long tail and one isolated point (117M). A sensible order: (1) inspect the 117M row and fix it if it is an error; (2) log-transform to handle the tail; (3) cap if the isolated point still stands out afterwards. **Count columns** (DEF_30 at 34, DEF_60 at 24, OBS_30 at 348, AMT_REQ_CREDIT_BUREAU_QRT at 261) are zero piles with an isolated point: a flag for “any or none”, plus a hand-picked cap.

Check yourself (answers in Appendix C)

- Why must you remove a placeholder before transforming the column?
- When would you cap instead of log-transform?
- Name two transforms that can handle zeros.
- What is the difference between capping and winsorising?
- Which tool fits a zero pile, and why does a log not fix it?

6 · When to deal with them, and when not to

6.1 Depends on the model

Model	Feature outliers	Skewed features	Notes
Logistic / linear regression	Fix errors; cap or transform extremes	Transform if |skew| is above about 1	Also check linearity of the effect
Regularised linear (ridge, lasso)	Same as above, plus scale features	Transform	Penalties assume similar feature scales
Neural network	Fix errors; scale; cap	Transform	Large inputs destabilise training
k-NN, SVM, clustering	Fix errors; scale robustly	Transform	Distances need comparable scales
Decision tree, random forest	Fix errors only	Usually leave	Ranking is all that matters
Gradient boosting (XGBoost, LightGBM)	Fix errors only	Usually leave	Transforming the target can still help

KEY IDEA

Errors and placeholders are fixed for **every** model. Transforms and caps are for models that care about the **numbers**. Since you will likely try both linear and tree models, prepare the transforms in a way you can switch on and off.

6.2 When to act, and when to wait

Act now (during EDA)	Wait (decide in modelling)
Impossible values and placeholders: log the rule and the number of rows	Whether to cap or transform a genuine long tail
Cross-column contradictions (e.g. family members fewer than children)	Which transform (log, Yeo-Johnson, rank)
Values you can prove are errors	Whether to drop a column with extreme but real values
Writing down every decision in your log	Whether the transform improves validation results

6.3 Order of operations and leakage

- **1.** Split into train and validation (and test) first when anything is learned from the data.
- **2.** Fix errors and placeholders by **rules** that do not depend on the data (a fixed value, a logical limit).
- **3.** Learn the quantities for capping, Box-Cox lambda, and scaling from the **training set only**.
- **4.** Apply those same numbers to validation and test.
- **5.** Keep the original column next to the transformed one, so you can compare.

WATCH OUT

Leakage. If you compute p99 or lambda on the full dataset, information from validation rows leaks into training. It makes results look better than they will be in real use. Rule-based fixes (like setting 365243 to NaN) are safe on all rows.

WATCH OUT

Do not fix by row number. “Set row 12840 to NaN” will not work on new data. State the rule on the value: *income above 50,000,000 → NaN*. Then the same rule runs on any future data.

6.4 How to decide that a fix helped

- The skew number moves into a lower band and the plot looks balanced.
- No new spike or gap appeared (a transform can create or reveal one).
- Validation performance does not get worse, and for linear models, usually improves.
- You can explain the fix in one sentence a colleague would accept.

6.5 In which stage does the data actually change?

EDA does not change the data. It records **plans**. The data changes in preprocessing, after you split into train and test. Keep the raw file untouched and make every change in re-runnable code.

Stage	What happens	Changes data?
1. Load	Read the raw file	No
2. EDA	Look, measure, write the log	No (plans only)
3. Fix known errors	Rules that learn nothing from the data	Yes, small
4. Train/test split	Hold out test data (stratify on the target)	No
5. Preprocessing	Impute, cap, transform, scale, encode, add flags; fitted on train only	Yes, the main step
6. Modelling and evaluation	Fit on processed train; apply the same fitted steps to test	No

Plan in the log	Stage
Set 365243 to NaN and add a flag; convert DAYS_ columns to years	3 · fix now
“Zero or not” flags (no learning needed)	3 or 5
Log transform of income, credit, annuity, goods price	5
Cap or winsorise isolated points (DEF_30, OBS_30, QRT, income)	5 · limits from train
Imputation and missing indicators (housing columns, OWN_CAR_AGE)	5 · fitted on train
Drop redundant columns (_MODE / _MEDI, one REGION_RATING)	5 · after the correlation check

KEY IDEA

Anything that **learns a number** from the data (median for imputing, percentiles for capping, mean and standard deviation for scaling) is fitted on train and then applied to test. A log transform has no learned number, so it cannot leak by itself.

Check yourself (answers in Appendix C)

- For which model family is a log transform of a feature most often unnecessary?
- Why compute the cap limits on the training set only?
- Why is “set row 12840 to NaN” a poor fix?
- In which stage do you set 365243 to NaN, and in which do you cap a column?

7 · A workflow for your project

7.1 Step by step

Step	What to do	Output
1. Summary table	For each numeric column: n_unique, missing %, zeros %, min, median, max, skew, abs_skew; sort by abs_skew	One table
2. Classify the columns	Flags and identifiers: skip. Few distinct values: bar chart. Bounded 0 to 1: plain histogram. Positive money: log x. Days: years	Plot lists
3. Check the lists	Overlap between lists should be empty; every numeric column must be in a plot list or a skip list	Clean lists
4. Plot by group	One figure per group of 6 to 9 columns; read each panel with the same questions	Figures
5. Record	Shape, whether a log helps, any anomaly, the decision, and the reason	Univariate log
6. Act	Fix errors by rule; flag placeholders; decide transforms later in modelling	Cleaned columns

7.2 Which plot for which column

What the table says	Plot
Continuous, moderate skew	Histogram with a box plot
Continuous, strong right skew, positive (money)	Histogram on a log x-axis (log-spaced bins)
Many zeros, or bounded 0 to 1	Histogram, log y-axis; note the zero share
Integer with more than about 20 distinct values (OBS_30/60)	Histogram with integer bins and a log y-axis
Few distinct values (counts under about 20, ratings, hour of day)	Bar chart of value counts
Negative day counts (DAYS_*)	Convert to years, then a histogram (exclude 365243 first)
Two-valued flags, constants, identifiers	No plot; value counts or drop
Look-alike groups (housing _AVG, _MODE, _MEDI)	Plot one per base measure after checking the correlation

IN YOUR LOAN-DEFAULT PROJECT

Your plot groups. Log x: AMT_INCOME_TOTAL, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE, REGION_POPULATION_RELATIVE, OWN_CAR_AGE. Log y with integer bins: OBS_30/60_CNT_SOCIAL_CIRCLE, AMT_REQ_CREDIT_BUREAU_YEAR. Histogram with box plot: EXT_SOURCE_1/2/3. Bar charts: CNT_CHILDREN, CNT_FAM_MEMBERS, DEF_30/60, AMT_REQ_CREDIT_BUREAU_HOUR/DAY/WEEK/MON/QRT, HOUR_APPR_PROCESS_START, REGION_RATING_CLIENT. Days to years: DAYS_EMPLOYED, DAYS_BIRTH, DAYS_ID_PUBLISH, DAYS_REGISTRATION, DAYS_LAST_PHONE_CHANGE.

7.3 Questions to ask of every panel

- Is it close to bell-shaped after the log, or still lopsided? On which side?
- Are there spikes at round values, or gaps? (Loan products create spikes at round amounts: real, not errors.)
- Does the tail look sensible after the outlier fixes?
- How many rows did the plot use compared with the non-missing count? (The filter x > 0 drops zeros.)
- Is it a near-duplicate of another column? (AMT_CREDIT and AMT_GOODS_PRICE look alike.)

A line for your log, per column: shape · does a log help (yes or no) · anomaly · decision · reason.

Column	Shape	Log helps?	Anomaly	Decision
AMT_CREDIT	Right-skewed, skew 1.2, spikes at round amounts	Yes (skew drops to about −0.3)	None	Log-transform for linear models
AMT_INCOME_TOTAL	Skew 391, one value of 117M	Yes (skew 0.17 after log)	117M is 6.5x the next	Set to NaN by rule, then log
DAYS_EMPLOYED	Spike at 365243	n/a	55,374 placeholder rows	NaN by rule, flag, then years
EXT_SOURCE_2	Moderately left-skewed (−0.79), bounded 0 to 1	No	None	Leave; no transform needed

8 · Common mistakes

Mistake	Why it hurts	Better
Deleting every point beyond 1.5 × IQR	On a long tail this removes thousands of real rows	Transform first, then apply the rule
Treating a placeholder as an outlier and capping it	The fake value is still in the column, just smaller	Set to NaN by rule, add a flag
Fixing by row number	Cannot be reproduced on new data	State the rule on the value
Log transform on a column with zeros or negatives	log(0) is undefined; rows vanish or errors appear	log1p, Yeo-Johnson, or log y-axis
Log axis with equal-width bins	Uneven bars and a cramped left side	np.logspace bins
Fitting cap limits on all the data	Leaks validation information	Learn on train only
Transforming for a tree model by habit	No gain, more work, harder to explain	Transform for the models that need it
Trusting the skew number alone	One point can inflate it; spikes mislead it	Check the plot and mean vs median
Reading skew as a low number when negative	−0.9 is moderate, not small	Use the absolute value
Plotting every column	Hours spent, nothing learned	Plot groups; skip columns the table already explains
Not recording the decision	Cannot explain it later	One line per column in the log

Appendix A · Code snippets

Each snippet assumes **s** is a numeric Series with missing values removed and **ax** is a matplotlib axis.

Summary table

```
rows = []
for col in numeric_cols:
    s = df[col].dropna()
    rows.append({"col": col, "n_unique": s.nunique(), "missing_%": 100*df[col].isna().mean(),
                 "zeros_%": 100*(s == 0).mean(), "min": s.min(), "median": s.median(),
                 "max": s.max(), "skew": s.skew()})
num_summary = pd.DataFrame(rows)
num_summary["abs_skew"] = num_summary["skew"].abs()
num_summary = num_summary.sort_values("abs_skew", ascending=False)
num_summary["skew_band"] = pd.cut(num_summary["abs_skew"], [0, 0.5, 1, np.inf],
                                  labels=["symmetric", "moderate", "strong"])
```

Type 1: log x-axis, log-spaced bins

```
s = s[s > 0]
bins = np.logspace(np.log10(s.min()), np.log10(s.max()), 40)
ax.hist(s, bins=bins)
ax.set_xscale("log")
```

Type 2: log y-axis (integer bins for counts)

```
bins = np.arange(s.min(), s.max() + 2) - 0.5
ax.hist(s, bins=bins)
ax.set_yscale("log")
```

Type 3: histogram of transformed values

```
ax.hist(np.log10(s[s > 0]), bins=40)     # positive values only
ax.hist(np.log1p(s), bins=40)            # keeps zeros
```

Types 4 and 5

```
ax.hist(s[s > 0], bins=bins); ax.set_xscale("log"); ax.set_yscale("log")      # log-log
ax.hist(s, bins=60); ax.set_xscale("symlog", linthresh=1)                     # symlog
```

Flags and fixes

```
q1, q3 = s.quantile([0.25, 0.75]); iqr = q3 - q1
flagged = (s < q1 - 1.5*iqr) | (s > q3 + 1.5*iqr)              # IQR rule
med = s.median(); mad = (s - med).abs().median()
robust_z = 0.6745 * (s - med) / mad                            # robust z-score
df["DAYS_EMPLOYED_FLAG"] = (df["DAYS_EMPLOYED"] == 365243).astype(int)
df["DAYS_EMPLOYED"] = df["DAYS_EMPLOYED"].replace(365243, np.nan)  # rule on the value
lo, hi = train[col].quantile([0.01, 0.99])                     # limits from TRAIN only
df[col + "_capped"] = df[col].clip(lo, hi)
```

Transformers that learn from data (fit on train, apply to all)

```
from sklearn.preprocessing import PowerTransformer, QuantileTransformer
pt = PowerTransformer(method="yeo-johnson").fit(train[[col]])
train[col + "_yj"] = pt.transform(train[[col]]); test[col + "_yj"] = pt.transform(test[[col]])
```

Appendix B · Glossary

Term	Meaning
Skew	Asymmetry of a distribution. Positive: long right tail. Negative: long left tail.
Outlier	A value unusually far from the rest, judged against the spread of the other values.
Long tail	Values thin out gradually toward one side; usually genuine data.
Isolated outlier	One or few points separated from the rest by a large gap; often an error.
Placeholder (sentinel)	A fake value standing for “unknown”, such as 365243 days.
Median, mean	Middle value; arithmetic average. The mean is pulled toward the tail.
Quantile, percentile	A value below which a given share of the data lies (p99 = 99%).
IQR	Q3 − Q1, the span of the middle half of the data.
MAD	Median absolute deviation; a robust measure of spread.
z-score	Distance from the mean in standard deviations.
Robust statistic	One that hardly changes when a few values are extreme (median, IQR).
Masking	Outliers inflating the standard deviation so much that their own z-scores look ordinary.
Log scale / log transform	Spacing or compressing values by ratio instead of by difference.
log1p	log(1 + x); handles zeros.
Box-Cox / Yeo-Johnson	Power transforms that choose lambda automatically; Yeo-Johnson also allows zeros and negatives.
Capping (clipping)	Replacing values beyond a chosen limit by that limit.
Winsorising	Capping where the limits are percentiles of the data (for example p1 and p99).
Zero pile	A spike at exactly 0; handled with a zero-or-not flag, not a log.
Leverage	How much a single point can pull a fitted line.
QQ plot	Data quantiles against theoretical quantiles; a straight line means a match.
Leakage	Learning anything from validation or test data during preparation.

Appendix C · Answers to “Check yourself”

Section 1

1. The mean, because it is dragged toward the tail.
2. Look at the ratios between the top values. A big jump from the largest to the second-largest (117M vs 18M) means isolated. Small steps mean a long tail.
3. Skew with no outliers: a smooth income-like long tail. An outlier that is not a long tail: one typo (age 900) in an otherwise symmetric column.

Section 2

1. The mean uses every value, so the extreme value adds a large amount to the sum. The median depends only on the middle position, which barely moves.
2. Tree-based models, because they split on the order of values, not their size.
3. The minimum and maximum define the range, so one extreme value stretches the range and compresses the rest into a tiny interval.

Section 3

1. Moderate (0.5 to 1). The tail is on the left, because the sign is negative.
2. The outliers inflate the standard deviation, which reduces their own z-scores.
3. On a long tail the rule flags many ordinary large values. After a log the shape is closer to symmetric, so the fences make sense.

Section 4

1. x-axis: the values (in real units); y-axis: counts. The bins are log-spaced, so each covers the same ratio, and the axis is drawn on a log scale.
2. The column contains zeros, and log(0) is undefined. Use a plain histogram with a log y-axis and integer bins.
3. A heavier right tail than the normal distribution (right skew).

Section 5

1. A placeholder is a fake value. A transform would only shrink it, not remove it, and it would still distort the shape and any statistics.
2. When the extreme values are real and must stay in the data, but a few would dominate a linear or distance-based model, and you want to keep the original scale.
3. log1p and Yeo-Johnson (also the square root, for non-negative values).
4. Both replace values beyond a limit by the limit. Capping uses a limit you pick; winsorising uses a percentile of the data (learned on train).
5. A zero-or-not flag. A log keeps the zeros at zero, so the spike stays.

Section 6

1. Tree-based models (decision tree, random forest, gradient boosting).
2. Otherwise information from the validation or test data leaks into the training process and results look better than they are.
3. It is tied to one row of one dataset. New data will not be fixed. A rule on the value works anywhere.
4. Setting 365243 to NaN is a stage-3 rule fix (before the split). Capping is a stage-5 preprocessing step with limits learned on train.

Appendix D · One-page cheat sheet

Question	Quick answer
Is it skewed?	|skew| below 0.5: no. 0.5 to 1: moderate. Above 1: strong. Confirm with mean vs median and a plot.
Is it an outlier?	Large ratio to the next value, beyond the IQR fences, or a robust z above 3, then check meaning.
Error or genuine?	Impossible or placeholder: error. Smooth tail from a real process: genuine.
Which plot?	Positive tail: log x. Spike at 0/1 or many zeros: log y. Few values: bars. Days: years.
Which fix?	Error: NaN by rule (+ flag). Genuine: log / Yeo-Johnson / cap for linear, nets, k-NN. Trees: leave.
When?	Errors now; transforms in modelling, learned on train only.
Always	Keep the original column and write the decision and the reason in your log.